
Investigating the Impact of REPA Loss on Training Speed and Generation Quality in Diffusion Models

Evgenii Pakhomov
HSE University

Dmitrii Gromak
HSE University

Sergei Kopylov
HSE University

Miron Makhlin
HSE University

Vladimir Novikov
HSE University

Abstract

Diffusion models have emerged as the dominant paradigm for high-fidelity image generation, yet their training remains computationally expensive and slow to converge. Recent work introduced REPA (REPresentation Alignment), a regularization technique that aligns projections of noisy input hidden states in diffusion transformers with clean image representations from pretrained visual encoders. Reported as an ICLR 2025 Oral, REPA demonstrated substantial improvements in training efficiency, achieving up to 17.5x faster convergence on large-scale ImageNet experiments, while also improving final generation quality. In this project, we aim to present a partial reproduction and systematic extension of REPA under constrained computational settings. Rather than scaling to full ImageNet, we plan to reproduce the core REPA framework on reduced datasets and smaller model configurations, verifying its key claim of accelerated convergence relative to standard diffusion training. Building on this, we plan to investigate REPA in low-data regimes, considering both subsampled large-scale datasets (e.g., ImageNet) and naturally small, domain-specific datasets such as Stanford Cars. We will study three main questions: (1) whether REPA’s improvements in convergence speed and sample quality persist when training data is limited; (2) how the choice of pretrained feature extractor – comparing alternatives such as DINOv2 and CLIP – affects performance, particularly in narrow visual domains; and (3) how sensitive REPA is to dataset scale, including whether its alignment objective can partially compensate for reduced data availability. Through these experiments, we aim to better characterize the practical benefits and limitations of representation alignment for efficient diffusion training in realistic, resource-constrained settings.

1 Introduction

Diffusion models have become one of the main paradigms for high-quality image generation, with successful architectures ranging from U-Net denoisers to latent diffusion models and diffusion transformers [16, 4, 15, 13]. However, training such models remains computationally expensive: not only must the model learn the denoising dynamics, but it must also be able to extract useful internal visual representations (discriminative features) in its hidden states [8, 20, 1, 11]

Recent work on REPA [21] proposes to accelerate diffusion training by explicitly aligning intermediate hidden states of the denoising model with clean image representations extracted by a frozen pretrained visual encoder. This suggests a simple but powerful idea: instead of forcing the diffusion model to discover good internal representations only through the denoising loss, we can use an external representation model as an auxiliary teacher.

In this project, we study representation alignment as a training principle for efficient diffusion modeling under constrained computational settings. We focus on reduced datasets, low-data regimes, and fine-grained visual domains. At first glance, one might expect REPA to be especially useful in such settings, since the pretrained teacher can inject visual knowledge that is unavailable from the small training set alone. However, this advantage is not obvious. Recent evidence from iREPA [18] suggests that the effectiveness of REPA is determined not merely by the global quality of the teacher encoder, but by the spatial structure of its features. Therefore, in narrow domains, a teacher encoder

may provide globally meaningful but locally mismatched representations, which can make the alignment objective less helpful or even harmful.

Our goal is to characterize when REPA improves convergence speed and sample quality, how its effectiveness depends on the pretrained feature extractor and dataset scale, and under which conditions the alignment objective can become a limitation rather than a reliable source of inductive bias. We also will try to verify the usefulness of REPA loss for text-diffusion models, as recently shown by [6].

2 Related work

Early attempts to bring representation information into diffusion models were mostly indirect. One line of work sought to make diffusion models themselves stronger representation learners, for example by reformulating score matching for unsupervised representation learning, sharing generative and discriminative parameters in joint diffusion models, or imposing bottlenecks and denoising redesigns that encourage semantically meaningful features [10, 2, 5, 1].

Some researchers tried injecting external semantic representations through multi-stage pipelines rather than direct hidden-state matching: Wuerstchen [14] learns a compact semantic image latent that guides diffusion, while Representation-Conditioned Generation [9] first generates self-supervised semantic codes and then conditions image synthesis on them.

REPA differs by making alignment explicit inside a single diffusion model, matching noisy intermediate denoiser features to clean pretrained visual embeddings and using this match as an auxiliary training signal.

iREPA further studies why some teacher encoders are more effective for representation alignment than others. Its main conclusion is that global semantic strength, such as classification accuracy, is not sufficient to explain the benefit of REPA. Instead, spatial structure in patch-level representations appears to be more important for image generation. This is especially relevant for our project, since fine-grained and narrow-domain datasets may require local details that are not well captured by a generic teacher encoder.

Subsequent work has studied how this idea should be adapted across architectures rather than transferred unchanged. In particular, U-REPA [19] shows that moving REPA to diffusion U-Nets is nontrivial because skip connections, stage-specific block roles, and spatial downsampling make transformer-style tokenwise alignment less effective; to address this, it aligns middle-stage U-Net features, upsamples them after projection, and adds a manifold loss to better bridge U-Net and ViT feature spaces.

REPA-E [7] extends the same principle to end-to-end latent diffusion transformers by jointly tuning the VAE tokenizer and the diffusion model, showing that representation alignment can stabilize a setting where diffusion loss alone is ineffective.

More recently, REPA-style alignment has also been adapted to masked diffusion language models [6], suggesting that representation alignment is evolving from a transformer-specific technique into a broader auxiliary training principle across generative architectures and even modalities.

3 Task setting

We follow the notation of REPA. Let $x \sim \mathcal{D}$ be a clean image from the training distribution, and let

$$z = E(x)$$

be its latent representation produced by the encoder E of a pretrained autoencoder. The base diffusion model follows the SiT velocity-prediction formulation used in REPA. Given Gaussian noise $\epsilon \sim \mathcal{N}(0, I)$, we define an interpolation path

$$z_t = \alpha_t z + \sigma_t \epsilon,$$

where $t \in [0, 1]$ and α_t, σ_t determine how the clean latent is gradually transformed into noise. The training target is the velocity of this path with respect to t :

$$u_t(z, \epsilon) = \frac{dz_t}{dt} = \frac{d\alpha_t}{dt} z + \frac{d\sigma_t}{dt} \epsilon.$$

The velocity model $v_\theta(z_t, t)$ is trained to approximate this target:

$$\mathcal{L}_{\text{diff}}(\theta) = \mathbb{E}_{z, \epsilon, t} \left[\|v_\theta(z_t, t) - u_t(z, \epsilon)\|_2^2 \right].$$

For the linear interpolation commonly used in this setting, $\alpha_t = 1 - t$ and $\sigma_t = t$, so $u_t(z, \epsilon) = \epsilon - z$.

89 REPA adds an auxiliary representation-alignment objective. Let f be a frozen pretrained visual
90 encoder, and let

$$y = f(x) \in \mathbb{R}^{N \times C}$$

91 be the clean image representation extracted by this encoder, where N is the number of patches and C
92 is the representation dimension. Let $H_\theta^\ell(z_t, t)$ denote the intermediate hidden states of the diffusion
93 transformer at layer ℓ . A trainable projection head h_ψ maps these hidden states into the teacher
94 representation space.

95 The REPA loss aligns projected noisy hidden states with clean teacher features by maximizing
96 patch-wise similarity:

$$\mathcal{L}_{\text{REPA}}(\theta, \psi) = -\mathbb{E}_{x, \epsilon, t} \left[\frac{1}{N} \sum_{n=1}^N \text{sim}(y_n, h_\psi(H_\theta^\ell(z_t, t))_n) \right],$$

97 where n indexes spatial patches and $\text{sim}(\cdot, \cdot)$ denotes the similarity function used for representation
98 alignment.

99 The final objective is

$$\mathcal{L}(\theta, \psi) = \mathcal{L}_{\text{diff}}(\theta) + \lambda \mathcal{L}_{\text{REPA}}(\theta, \psi),$$

100 where λ controls the strength of the alignment term.

101 Our main experimental question is whether this auxiliary alignment term improves training in
102 small-scale and narrow-domain settings. We will compare the vanilla diffusion objective against
103 REPA-style training under matched architecture, optimizer, data, and compute budgets. We will
104 also vary the frozen representation encoder f and study whether its representations provide useful
105 guidance for generation.

106 This setting allows us to test two competing hypotheses. The optimistic hypothesis is that REPA
107 should be especially useful in low-data regimes because the pretrained encoder supplies an external
108 representation prior that the diffusion model would otherwise need to learn from limited data. The
109 skeptical hypothesis is that this prior may be mismatched. In particular, a teacher encoder can encode
110 strong global semantic information while still having patch-level representations that are poorly
111 aligned with the spatial structure required in a narrow domain. In that case, minimizing $\mathcal{L}_{\text{REPA}}$ may
112 force the diffusion model to match representations that are globally meaningful but locally suboptimal
113 for generation, reducing or even reversing the expected benefit of REPA.

114 4 Metrics and Evaluation

115 We evaluate models along three main axes.

116 **Generation quality.** The primary metric for image generation is Fréchet Inception Distance
117 (FID) [3], which measures the distributional gap between generated and real images in the feature
118 space of a pretrained Inception-v3 network. Lower FID indicates better quality and diversity together.
119 We will complement FID with qualitative inspection of random samples, since models with similar
120 FID scores can differ noticeably in perceptual quality – especially on fine-grained domains where
121 structural artifacts may not be captured by Inception features. For Stanford Cars, we will additionally
122 assess part-level coherence, texture realism, and shape consistency. We will also assess diversity,
123 which is important for image generation. For this we will use the mean pairwise LPIPS distance over
124 a set of randomly sampled image pairs from the generator:

$$\text{diversity} \approx \frac{1}{K} \sum_{\substack{i, j=1 \\ i \neq j}}^K \text{LPIPS}(x_i, x_j),$$

125 where higher values indicate greater perceptual spread across samples. We will additionally inspect
126 samples visually for mode collapse.

127 **Training efficiency.** Following the evaluation protocol of the original REPA paper [21], we will
128 plot FID as a function of both training steps and wall-clock time, evaluating checkpoints at fixed
129 intervals (e.g., every 20k steps). This will allow us to compare convergence speed independently
130 from final quality: REPA’s core promise is that alignment accelerates the early training phase, so a
131 model with REPA should reach a target FID threshold significantly faster than the baseline. We will
132 report both the step count and GPU-hours required to reach representative FID thresholds as scalar
133 efficiency indicators.

Representation behavior. To understand *why* REPA helps or fails in a given setting, we will try to track several internal diagnostics throughout training. For example, we plan to log the alignment loss $\mathcal{L}_{\text{REPA}}$ alongside the diffusion loss $\mathcal{L}_{\text{diff}}$, as well as monitor the mean cosine similarity between projected hidden states $h_{\psi}(H_{\theta}^{\ell})$ and teacher patch embeddings y . We will also measure linear probing accuracy on the intermediate hidden states of the diffusion model (frozen at various checkpoints) as an indicator of whether the denoiser is learning discriminatively useful representations. Together, these diagnostics help distinguish settings where REPA successfully transfers teacher representations from settings where alignment is achieved superficially without improving generation.

Text diffusion metrics (if time allows). For the masked text diffusion extension, we will use perplexity on a held-out validation set as the primary metric, following [6, 17]. We will also track the alignment loss and cosine similarity between hidden states and teacher token embeddings to verify that the alignment objective is active and meaningful during training.

5 Plan

Step 1: Baseline reproduction. We plan to begin by implementing a self-contained REPA training pipeline on a reduced subset of ImageNet (10% of classes, $\approx 130\text{k}$ images) at 64×64 resolution, using a SiT-S/2 backbone [21] and DINOv2-B [12] as the frozen teacher. We will train two models under identical conditions – one with the REPA loss and one without – for 400k steps and compare their FID-vs-step curves. Success here means reproducing the qualitative trend of faster convergence with REPA, even if the absolute FID numbers differ from the full-scale result. This phase will validate our infrastructure and give us a reliable baseline for subsequent ablations.

Step 2: Low-data and fine-grained domain study. We plan to investigate two complementary low-data regimes. First, we will subsample ImageNet at four data fractions (1%, 5%, 10%, 20%) and train REPA vs. baseline models to the same step budget, measuring how FID and convergence speed scale with dataset size. We hypothesize that REPA’s benefit is largest at moderate data regimes (5–20%) where the encoder prior can compensate for limited data, but may saturate at very low fractions where neither model can generate coherent images. Second, we will train on Stanford Cars ($\approx 8\text{k}$ images), a fine-grained benchmark where object geometry and part structure matter more than global semantics. This will test the hypothesis from iREPA [18] that teachers with strong global features but weak spatial structure (e.g., CLIP) may be less helpful – or even harmful – in domains that require precise local detail.

Step 3: Teacher encoder ablation. We plan to compare three pretrained visual encoders as REPA teachers on both the subsampled ImageNet (10%) and Stanford Cars settings: DINOv2-B [12] (strong spatial structure, patch-level features), CLIP ViT-B/32 (strong global semantics, weaker patch structure), and a supervised ResNet-50 (conventional discriminative baseline with no patch-level design). For each teacher, we will measure FID, the alignment loss trajectory, and the linear probing accuracy of the aligned hidden states. This ablation will directly test iREPA’s spatial-structure hypothesis in our constrained settings, and determines whether teacher choice should depend on dataset domain.

Step 4: Text diffusion extension. Motivated by [6], who showed that REPA accelerates training of Portuguese Masked Diffusion Language Models (MDLM [17]) by 28.6% in perplexity and confirmed that mid-level alignment with a modern language encoder works best, we plan to adapt this finding to English-language masked diffusion. Concretely, we want to train a small MDLM on a text corpus (e.g., WikiText-103) with and without a REPA-style alignment loss. Following [6], we will align intermediate transformer blocks of the MDLM with token representations from a frozen pretrained encoder. We will compare two teacher choices: a bidirectional encoder (BERT-based) and a decoder-based model (GPT-based), mirroring the BERTimbau vs. Qwen comparison from the original paper. We will evaluate perplexity on the validation set at fixed training step checkpoints to assess whether the convergence speedup observed for Portuguese generalizes to English and to a different model scale.

6 Expected outcomes

The central output of the project is an empirical characterization of when and why REPA works under constrained settings. For the image experiments, we expect REPA’s convergence benefit to be robust at moderate data scales (5–20% of ImageNet) but to degrade in very low-data regimes, where the alignment signal becomes too weak relative to the denoising objective to provide a reliable training prior. On fine-grained domains such as Stanford Cars, we expect teacher choice to matter substantially: teachers with strong spatial patch structure (DINOv2) should outperform those with

190 primarily global semantic features (CLIP), consistent with the findings of iREPA [18]. More broadly,
191 we may find that a teacher which ignores domain-specific visual features can actively harm training
192 by pulling hidden states toward representations that are globally coherent but locally mismatched to
193 the target distribution.

194 For the text diffusion extension, we expect the qualitative finding of [6] to generalize beyond
195 Portuguese: mid-level alignment with a bidirectional encoder should reduce perplexity and accelerate
196 convergence compared to the unaligned baseline, while decoder-based teachers may prove less
197 effective due to their weaker bidirectional token structure. A negative or null result would itself be
198 informative, suggesting that the REPA benefit in text diffusion is tied to specific linguistic or resource
199 conditions rather than being a general property of representation alignment.

200 Taken together, the project should clarify whether REPA is best understood as a universal training
201 accelerator or as a representation prior whose practical benefit depends jointly on the teacher’s feature
202 structure, the scale and domain of the training data, and the alignment design choices. If REPA proves
203 fragile across these axes, this would motivate follow-up work on adaptive or domain-aware alignment
204 strategies.



Figure 1: Memorization examples on Imagenette and CelebA. See full examples in Appendix A.

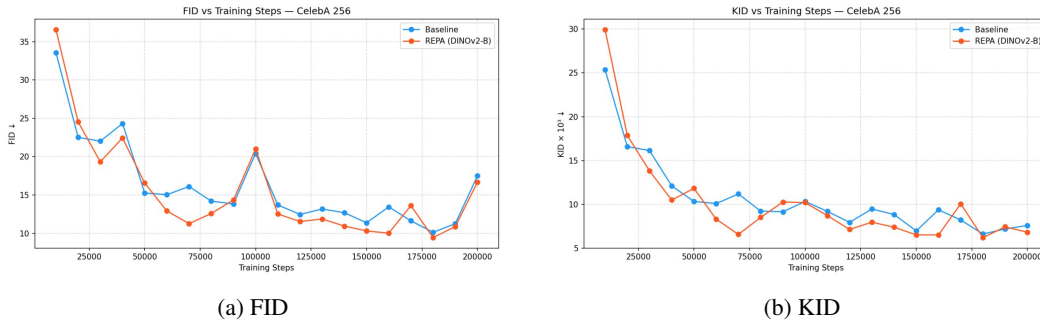


Figure 2: FID and KID plots for training checkpoints

7 Baseline Reproduction

We begin by reproducing the key results of the original REPA paper: faster convergence in terms of FID and improved stability of model features during training.

7.1 Imagenette

We first train on Imagenette – a subset of ImageNet containing 10 easily classified classes with approximately 1 000 samples per class. After 100 000 iterations, the model achieves good generation quality, but at a cost: the model is sufficiently large to memorize the entire dataset, a tendency amplified by the REPA loss, as shown in Figure 1a. This reveals a key risk of applying diffusion models to small-scale datasets – complete training set memorization rather than genuine generalization.

We address this problem by [MIRON].

7.2 CelebA

Since training on small, multi-domain datasets leads to memorization, we investigate whether a single-domain dataset can mitigate this issue. We select the Large-scale CelebFaces Attributes (CelebA) dataset as our fine-grained benchmark and train the diffusion model on it. Notably, the memorization problem does not arise here, as illustrated in Figure 1b, enabling analysis analogous to that in the original REPA paper.

7.2.1 Training Speedup Analysis

Figure 2 shows FID and KID as a function of training iteration. Both models improve steadily in generation quality. Early in training, the REPA model exhibits higher total loss, consistent with the original paper’s observation that the alignment term adds overhead while the model is still learning to extract meaningful features. By iteration 30 000 this gap closes, and by iteration 70 000 the REPA model reaches $\text{FID} \approx 11$ – a threshold the baseline only reaches at iteration 180 000. This corresponds to a $2.5\times$ speedup in early training, corroborated by the KID curve, where the REPA model also attains its all-time minimum at iteration 70 000.

Beyond this point, both models’ metrics degrade at a similar rate. Once $\mathcal{L}_{\text{REPA}}$ approaches zero around iteration 70 000, the remaining gradient signal comes entirely from the diffusion loss, causing

the two training curves to converge. The periodic quality dips visible in both curves likely reflect transient instabilities from a learning rate that is too high at later stages. It is also worth noting that these metrics are evaluated on the non-EMA model, which amplifies measurement noise.

We attribute the early saturation of REPA’s benefit to the relative simplicity of CelebA’s spatial structure: facial images share consistent geometry, pose, and composition, allowing the model to absorb the essential representational signal from the teacher within 70 000 iterations, after which training reduces to standard diffusion fine-tuning.

7.2.2 Structural Analysis

To move beyond scalar metrics such as FID and KID, we analyse the internal representations of both models directly by visualising the patch-level hidden states at each transformer block. This section describes how those visualisations are constructed and what they reveal about the effect of REPA on a face-generation task conditioned on facial attributes.

Construction of PCA Feature Maps. A SiT-B/2 transformer processes its input as a sequence of spatial patches. After the VAE encodes a 256×256 image into a $32 \times 32 \times 4$ latent z_0 , a patch embedding layer with patch size 2 splits this into $16 \times 16 = 256$ tokens, each carrying a 768-dimensional feature vector. At every transformer block $\ell \in \{1, \dots, 12\}$ the model produces a hidden state matrix

$$H^{(\ell)} \in \mathbb{R}^{256 \times 768}.$$

To visualise the geometry of this representation space, forward hooks are registered on every block simultaneously so that a *single* forward pass captures all 12 hidden states at once. The input to the model is a noisy latent constructed via the linear stochastic interpolant

$$z_t = (1 - t) z_0 + t \varepsilon, \quad \varepsilon \sim \mathcal{N}(0, I),$$

evaluated at three noise levels $t \in \{0.5, 0.7, 0.9\}$ to probe how representations change as the signal-to-noise ratio decreases.

At each block the 256 token vectors are projected to three dimensions via PCA:

$$\tilde{H}^{(\ell)} = \text{PCA}_3\left(H^{(\ell)}\right) \in \mathbb{R}^{256 \times 3}.$$

Each component is normalised channel-wise to $[0, 1]$, reshaped to a $16 \times 16 \times 3$ spatial grid, and upsampled to 128×128 pixels for display. Patches whose token vectors are similar appear in the same colour, so the map is a spatial portrait of what the model considers semantically equivalent at that depth.

A critical implementation detail is that the PCA for each layer is fitted *jointly* on the concatenated token matrices of the Baseline and REPA models:

$$\text{PCA}_3\left([H_{\text{Base}}^{(\ell)}; H_{\text{REPA}}^{(\ell)}]\right),$$

so both models share the same colour axes. Without this, a colour difference between the two maps could reflect an arbitrary rotation of the PCA solution rather than a genuine representational difference. The joint fitting guarantees that identical colours encode identical representations across models.

Layer-wise Progression and Noise Robustness. Figure 3 shows PCA feature maps for a representative image across all 12 transformer blocks and three noise levels $t \in \{0.5, 0.7, 0.9\}$. For each timestep, rows correspond to the original photograph, the Baseline PCA map, and the REPA PCA map respectively, with transformer blocks as columns.

Figure 3: PCA feature maps across all 12 transformer blocks and three noise levels. For each timestep group, rows show: the original CelebA image, the Baseline hidden-state PCA map, and the REPA hidden-state PCA map. Columns correspond to transformer blocks 1–12 (left to right).

In the *Baseline* model, PCA maps remain largely unstructured through the first five to six blocks. Patches from visually distinct face regions – hair, skin, eyes, background – are assigned similar colours, indicating that early token representations do not yet differentiate semantic content. Meaningful spatial organisation appears only in the final two or three blocks, and even there it is coarser than what REPA achieves.

In the *REPA* model, semantic structure emerges substantially earlier. Already by block 6, the PCA maps consistently partition the image into coherent face regions: a compact colour cluster for skin, a distinct cluster for hair, and a separate background region. This earlier formation is a direct consequence of representation alignment.

277 The timestep axis reveals the most striking contrast. At $t = 0.5$ (moderate noise) both models
 278 produce partially structured maps in the middle layers. At $t = 0.9$ (near-pure Gaussian noise) the
 279 Baseline maps collapse to spatially incoherent patterns at all but the deepest blocks, while REPA
 280 maps retain clear semantic region boundaries. This is particularly telling on CelebA: because all faces
 281 share nearly the same spatial layout, a model can recover coarse centre-face structure at high noise
 282 simply by memorising positional priors, without encoding any attribute-level content. The Baseline
 283 exploits this shortcut. REPA maps, anchored to DINOv2 features that encode attribute-level content,
 284 maintain finer internal structure – skin, hair boundary, eye region – even when the noisy input carries
 285 almost no pixel-level signal.

286 To conclude, Representation Alignment enables the model to learn coherent, meaningful features
 287 on early layers as well as makes these features more robust and stable so that they can be reliably
 288 extracted even from pictures with low pixel-level signal.

289 **Attribute-Specific Separability.** Our training setup conditions the model on 16 class labels derived
 290 from four binary attributes: *Male*, *Smiling*, *Young*, and *Attractive*. This allows us to assess whether
 291 REPA’s representational improvements are attribute-dependent – an analysis that goes beyond the
 292 original paper’s ImageNet evaluation.

293 Figure 4 shows representative image pairs for each attribute, selected automatically to satisfy three
 294 constraints. First, each pair is *controlled*: the two images belong to class labels that differ in exactly
 295 one attribute bit while all other three attributes are identical, isolating the effect of the target attribute.
 296 Second, pairs are required to be visually similar in terms of pixel-level MSE, controlling for pose,
 297 lighting, and background. Third, no image is reused across attribute panels. Among candidate pairs
 298 satisfying these constraints, we select the one that maximises the relative gain in mean patch-level
 299 cosine distance under REPA compared to the Baseline:

$$\text{score} = \frac{d_{\text{REPA}} - d_{\text{Baseline}}}{d_{\text{Baseline}} + \varepsilon}, \quad d(A, B) = 1 - \frac{1}{N} \sum_{n=1}^N \frac{A^{[n]} \cdot B^{[n]}}{\|A^{[n]}\| \|B^{[n]}\|}.$$

Figure 4: Attribute-specific feature map separability at block 8 and $t = 0.5$. The figure is arranged
 as a 2×2 grid of attribute panels (Male, Smiling, Young, Attractive). Within each panel, the two
 rows show images differing only in the target attribute (all other attributes held fixed); columns
 show the original photograph, the Baseline PCA map, and the REPA PCA map. Pairs were selected
 automatically to be visually similar and to maximise REPA’s relative token-space separation gain
 over Baseline.

300 The results exhibit a clear attribute-dependent hierarchy. *Male* shows the largest visual separation
 301 under REPA: colour clusters corresponding to hair, skin tone, and facial structure diverge markedly
 302 between the Male=N and Male=Y rows in the REPA maps, while the Baseline maps remain nearly
 303 indistinguishable. Gender is one of the strongest signals encoded by DINOv2, providing a rich teacher
 304 signal for the alignment loss. *Smiling* produces a spatially localised effect: a distinct patch cluster
 305 appears around the lower-face region for Smiling=Y images in the REPA maps, reflecting alignment
 306 of mouth-area tokens with DINOv2 features sensitive to facial expression; this cluster is absent in
 307 the Baseline. *Young* yields moderate gains, consistent with age-related textural information (skin
 308 smoothness, fine structure) being captured by DINOv2 but less saliently than gender or expression.
 309 *Attractive* shows the smallest improvement. The attribute is subjectively defined and inconsistently
 310 grounded in low-level visual features, so the DINOv2 teacher provides a weaker alignment signal
 311 and the representational gain is correspondingly smaller.

312 This hierarchy reflects a fundamental property of the REPA framework: it can only transfer
 313 structure that the self-supervised teacher itself encodes. The LDA separability of DINOv2 features
 314 with respect to each attribute therefore acts as an upper bound on the representational gain that REPA
 315 can deliver.

316 **Key Takeaways.** CelebA images are spatially homogeneous (face centred, consistent scale), and
 317 the class conditioning is injected at every transformer block via adaptive layer normalisation, making
 318 all 16 class labels trivially linearly decodable from even the earliest hidden states in both models. The
 319 PCA visualisations serve as the primary evidence for representational quality, since they reveal the
 320 *spatial organisation* of features rather than their mere linear separability.

321 The key takeaway is that REPA forces the model to internalise semantically meaningful facial
 322 structure earlier in the network and more robustly under noise, even though the generation objective
 323 alone would not require it on a spatially homogeneous dataset like CelebA.

324 7.2.3 VOVA, attribute direction vectors analysis, appendix C

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